SEAGAS - Team Update: What Is Done, What Is Left, What To Do After Pulling

From Carl (Soh Way Miin), Backend Developer, to Group 10 | 16 October 2026
COS40005 Computing Technology Project A | POC004 | Sprint 2 report due 24 October

In short: the website now works end to end with the NLP inside it, the security issues from the 16 October progress check are fixed, the advisor dashboard has real data, and admins can manage accounts, the skills library and job roles and read an audit log. After you pull, run **start_all.bat** once and sign in again (section 3).

1. What Is Done

1.1 This week's backend work (now on dev)

Area	What changed
NLP in the website	Pasted job descriptions are analysed by Yong Hin's NLP; assessments use his Strong / Developing / Gap matcher. Every skill shows a one-sentence reason (Job Analysis list, Skill Gap on hover). Job Analysis lets you choose Hybrid, Keyword (Level 1) or Semantic (Level 2).
Security	Students can only see their own assessments, recommendations and JDs. No default secret key; a random one is generated. Passwords 8 characters to 72 bytes. Profile, skill and project input is validated, so bad input gives a message instead of a server error.
Bug fixes	Deleting a skill works. Dashboard shows the 7 dimension scores. Project, certificate and internship scores change when you add records. Advisor Gap % counts each student once.
Batch assessments	`data/run_assessments.py` assesses every synthetic student against their target role (5,000 in about 30 s). Scores follow the data: strong 74.7%, average 56.0%, weak 32.4%. The start script runs it once, so the advisor charts are no longer empty.
Audit log	Sign-ins (also failed ones), registrations, assessments, JD analyses, advisors opening a student, and every admin change are recorded. Admins read it on the new Audit Log page. No passwords are stored in it.
Admin portal	User Management: add advisor/admin, edit name, reset password, (de)activate. Skills & Roles: add skills and roles, choose a role's required skills, switch any skill or role on or off.
Accounts and login	One login page; the role in the database opens the student, advisor or admin portal. Only students self-register. Emails: @student / @advisor / @admin.newinti.edu.my.
Tests and docs	`tests/test_backend_api.py` (35 API tests on a temporary database). README rewritten: setup, accounts, NLP install, tests.

1.2 Who owns the skills and roles data

The 138 skills and 6 roles from `data/skills/skills_library.csv` and the seed files are still maintained there (Yong Hin), and the start script re-applies them. In the Admin page they can only be switched on or off, and that choice survives the seeds. Skills and roles added in the Admin page (marked **admin**) are fully editable there and never touched by the seeds.

2. Project Status: the 13 Deliverables

9 of 13 are done or mostly done (2 on the morning of 16 October).

#	Deliverable	Status	Notes
1	Working web platform	**Mostly done**	Role template and pasted JD flows work; three portals; one login.
2	Student profile	**Partial**	Missing: evidence when adding a skill, work experience, activities.
3	JD NLP module	**Done**	Keyword, semantic, hybrid; connected.
4	AI skill matching	**Done**	Strong / Developing / Gap with reasons.
5	Readiness score	**Done**	Weighted; experience dimensions update.
6	Skill gap analysis	**Done**	New matcher; reasons shown.
7	Recommendation engine	**Done**	298 resources, beginner to advanced.
8	Student dashboard	**Done**	7 dimension scores load.
9	Advisor dashboard	**Mostly done**	Has cohort data now. Missing: review / notes on AI results (human oversight), report export page, softer "At Risk" wording.
10	Development tracking	**Done**	Start / complete / summary.
11	AI evaluation	**Partial**	65-JD method comparison done; 0 gold labels, so no precision / recall / F1.
12	Usability (SUS)	**Planned**	Tasks and participants not prepared.
13	Docs, report, demo	**Partial**	NLP docs and README done; Sprint 2 report due 24 Oct.

3. What To Do After You Pull

3.1 Every member

1. `git checkout dev` then `git pull`.
2. Double-click **start_all.bat** (or `backend\start_backend.bat` then `frontend\start_frontend.bat`). It updates your database by itself: new columns, the audit log table, the views, the recommendations library.
3. Watch the backend window. The first run after this update also:
   - replaces the old public secret key in `backend\.env` with a random one: **everyone has to sign in again**;
   - assesses the 5,000 synthetic students once (about 30 s to 1 min);
   - asks to create an admin if your database has none: the email must end with `@admin.newinti.edu.my`. You type the password; it is never stored in the code.
4. Open http://localhost:5173 and sign in. Test student: `student0001@synthetic.example.com` / `Seagas@2026`.
5. For an advisor account: sign in as admin, open **User Management**, **+ Add user**, role Advisor, email ending `@advisor.newinti.edu.my`.

3.2 Optional

- **Hybrid and semantic matching:** `python -m pip install -r nlp\requirements.txt` (about 1 GB, mostly torch), then restart the backend. The first start downloads the model (~90 MB) and shows "NLP model loaded (hybrid ready)" after 20-30 s. Without it the site uses keyword matching and shows a short notice.
- **Tests:** `python -m pip install -r tests\requirements.txt` then `python -m pytest tests -q`. The API tests create and delete their own database.

3.3 If something goes wrong

Problem	Fix
Backend window: "SECRET_KEY is not set"	Run `start_backend.bat` (it creates the key), or `python ensure_secret_key.py` in `backend`.
Logged out / "Invalid or expired token"	Expected once after the update (new key). Sign in again.
Frontend: EPERM ... node_modules\.vite\deps	OneDrive locks the Vite cache. Delete `frontend\node_modules\.vite` and start again; best is to keep the project outside OneDrive.
Do not know an advisor's password	Passwords are stored only as hashes. Sign in as admin and use **Reset password** in User Management.
Yellow notice "Hybrid matching is not available"	The semantic packages are not installed (3.2). Keyword results are still valid.
"No skills were found in this job description"	The JD names no concrete skills, or is very short. Paste the full advert; see the NLP known issues document.

4. What Is Still Left

Owner	Task	When
Everyone	Pull and run start_all.bat once (section 3); check your own pages in the browser	Now
Yong Hin	Gold labels (20-30 JDs, second annotator for Cohen's kappa) and precision / recall / F1; fix generic words and aliases ("testing" to A/B Testing, "web-development"); TF-IDF baseline; 30 JDs per role; update Sprint 2 backlog status and hours	Sprint 3-4
Ren Hang, Shanice	Evidence choice when adding a skill (the NLP evidence rule needs it); highlight extracted skills in the JD text; "decision support" notice on Dashboard and Skill Gap; "At Risk" to "Needs support"; show errors instead of empty catch blocks	Sprint 3
Aaron	Human oversight: advisor review and notes on AI results; career path API and page	Sprint 3
Carl	Advisor report export page with Ren Hang; help with evidence derivation from projects; support the integration and bug fixes	Sprint 3
Whole team	Decide NLP rule proposals A / B / C; Sprint 2 report by 24 Oct; SUS usability test; final report and demo	24 Oct / Sprint 4

Details for the NLP items are in `docs/SEAGAS_NLP_RemainingWork_16October.pdf`; today's backend changes are in `docs/SEAGAS_WorkReport_16October_By_Waymiin.pdf`.

Note: these changes were pushed straight to dev, without a pull request. Please look through the parts that touch your work and tell Carl about anything that looks wrong.